

Calibrated, Disagreement-Aware Stacked Ensembling with Stability-Weighted Consensus Feature Selection for Multi-Class Intrusion Detection on UNSW-NB15

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Abstract

Static-benchmark intrusion detection studies on UNSW-NB15 routinely report headline accuracies above **97%**, yet rarely audit the reliability of the underlying class-probability estimates or the behaviour of the ensemble under model disagreement. We present a leakage-safe, reliability-focused ensemble architecture that combines (i) per-base-learner out-of-fold isotonic/sigmoid calibration selected by Brier score, (ii) a reliability-criterion weighted average whose model weights jointly penalise log loss and expected calibration error, (iii) a multinomial-logistic stacker trained on full calibrated probability vectors and inter-model disagreement descriptors, and (iv) a sample-level disagreement-aware gate that regresses the closed-form optimal mixing coefficient $\beta^*(\mathbf{x})$ between the stacker and weighted-average paths. Feature inputs are chosen by a stability-weighted consensus selector that aggregates four heterogeneous selectors with a calibration-aware penalty. On the official UNSW-NB15 multi-class split with five-fold cross-validation on the training partition, the proposed system attains macro $F_1 = 0.659 \pm 0.018$, statistically significantly outperforming every single-model and naive-ensemble baseline (paired t -test $p < 0.05$ for all ten comparators; Wilcoxon $p = 0.0312$). On the locked test set the proposed system attains the best Brier score (**0.0306**), best macro ROC-AUC (**0.9475**) and best accuracy (**0.7685**); plain weighted averaging leads on test macro F_1 by **0.032**, an honest finding we attribute to the well-documented UNSW-NB15 train/test distribution shift and which we report rather than hide. We frame the contribution as a reliability-focused improvement, not a leaderboard claim, and release the full configuration, deterministic seed registry, and run-manifest artefacts to support exact reproduction.

Keywords: Intrusion detection, UNSW-NB15, Ensemble learning, Probability calibration, Disagreement-aware gating, Feature stability, Class imbalance

1 Introduction

Network intrusion detection systems (NIDS) on the UNSW-NB15 corpus [Moustafa and Slay(2015)] are most often evaluated through a single static train/test split with overall accuracy or macro F_1 as the headline metric, and the literature is now saturated with reports of 97–99% binary accuracy. Yet the reliability of the predicted class probabilities, the behaviour of stacked ensembles under base-model disagreement, and the leakage discipline of the evaluation protocol are rarely audited. Three weaknesses recur in recent UNSW-NB15 ensemble work: base-model probabilities are typically left uncalibrated even when downstream stackers consume them as features; meta-learners are routinely fed only the per-model maximum probability, discarding cross-class structure and inter-model disagreement signals that are most informative precisely when the ensemble is least confident; and fold discipline is inconsistent, with selectors and calibrators sometimes fitted on the full training set before cross-validation. This paper takes the opposite stance: rather than chase the leaderboard, we build a calibrated, disagreement-aware ensemble whose contribution lies in decision quality under class imbalance.

We close the gaps above with two concrete contributions. First, we propose an integrated calibrated-stacker + weighted-average + analytic- β^* gate architecture in which every component is fitted strictly inside fold boundaries on out-of-fold (OOF) probability data; to our knowledge no published UNSW-NB15 study combines all four ingredients in one pipeline. Second, we introduce a stability-weighted consensus feature selector that aggregates four heterogeneous selectors (ANOVA F-test, mutual information, Random Forest importance, SGD ElasticNet coefficients) under a joint criterion that multiplies inner-fold macro F_1 , pairwise-Jaccard stability, and a calibration-aware penalty ($1 - \text{ECE}$).

2 Related Work

Stacking, bagging and gradient-boosted ensembles dominate the recent UNSW-NB15 literature [Kasongo and Sun(2020), Zoghi and Serpen(2024), Suresh et al(2025)Suresh, Ramesh, and Anand]. Zoghi and Serpen [Zoghi and Serpen(2024)] combine Balanced Bagging, XGBoost and Random Forest with a Hellinger-distance criterion and threshold tuning, achieving 97.80% binary accuracy but reporting no calibration analysis, while Kasongo and Sun [Kasongo and Sun(2020)] compare feature-selection methods on UNSW-NB15 with XGBoost without examining the calibrated probability quality of the selected models. Recent IJIT contributions on intrusion detection [Kumar et al(2024), Sharma et al(2025), Bediya and Kumar(2020)] similarly emphasise accuracy and macro F_1 without auditing reliability.

On the calibration side, Platt scaling [Platt(1999)] and isotonic regression [Zadrozny and Elkan(2002)] remain the standard binary calibrators, and Niculescu-Mizil and Caruana [Niculescu-Mizil and Caruana(2005)] showed that boosted and bagged tree ensembles are systematically miscalibrated and benefit from post-hoc correction; for multi-class problems, Kull et al. [Kull et al(2019)Kull, Perelló-Nieto, Kängsepp, Silva Filho, Song, and Flach] proposed Dirichlet calibration, but we use the simpler one-vs-rest decomposition because it integrates cleanly with our OOF-fitting protocol. The methodological ancestor of our stacker path is Wolpert’s stacked generalisation [Wolpert(1992)], while the dynamic ensemble selection literature, exemplified by META-DES [Cruz et al(2015)Cruz, Sabourin, Cavalcanti, and Ren], learns competence on a per-instance basis from meta-features; our gate is closer in spirit to Mixture-of-Experts gating [Jacobs et al(1991)Jacobs, Jordan, Nowlan, and Hinton] in that it produces a continuous mixing coefficient between two pre-aggregated paths rather than selecting from a pool of base learners. Aggregating multiple selectors to improve feature-selection robustness is well-established [Wang and Feng(2017), Almomany et al(2022)], but our selector adds two ingredients absent from prior UNSW-NB15 work: a calibration-aware penalty in the inner-fold criterion, and an explicit pairwise-Jaccard stability weight. The components above are individually well-trodden; our novelty lies in their integrated assembly and in the closed-form analytic- β^* formulation of the gate’s training target.

3 Dataset and Preprocessing

We use the official UNSW-NB15 train/test partition published by the Australian Centre for Cyber Security [Moustafa and Slay(2015)]: 82 332 training samples and 175 341 test samples, with 42 feature columns (after removing the synthetic `id` column) and two label columns, `attack_cat` (10-class multi-class target) and `label` (binary). Class distributions are highly imbalanced; Table 1 summarises the per-class counts.

A frozen schema specification fixes the column order, dtypes, and the multi-class label vocabulary; the loader rejects any input that violates the schema, strips whitespace and BOMs introduced by upstream tooling, and guarantees that label columns are disjoint from the feature column set. Because a single shared transformer cannot satisfy the inductive biases of trees, linear models, and distance-based learners simultaneously, we implement three preprocessing contracts: a *tree* pipeline (median imputation, ordinal encoding with explicit unknown values, no scaling) for Random Forest, Extra Trees and XGBoost; a *linear* pipeline (median imputation, RobustScaler, one-hot encoding with rare-category grouping at `min_frequency=5`) for the stacker and linear selectors; and a *KNN* pipeline identical to the linear contract, since ordinal-coded categoricals must not be fed to a distance-based model. All transformers are fitted strictly on the training fold, with column order frozen at fit time and validated at transform time. To eliminate formula drift, every engineered feature is also centralised in an audited registry: each entry is a `FeatureSpec` with name, formula, input columns, category, and a clipping policy of either $[0, 1]$ (for ratio-valued features) or a train-fold-fitted percentile range. Table 2 lists the seventeen engineered features.

Table 1 UNSW-NB15 official-split class distribution.

Multi-class is the primary task; the binary task is used only as an internal sanity check ($> 98\%$ across all metrics) and is not pursued further in this paper.

Class	Train	Train %	Test	Test %
Normal	37 000	44.94	56 000	31.94
Generic	18 871	22.92	40 000	22.81
Exploits	11 132	13.52	33 393	19.04
Fuzzers	6 062	7.36	18 184	10.37
DoS	4 089	4.97	12 264	6.99
Reconnaissance	3 496	4.25	10 491	5.98
Analysis	677	0.82	2 000	1.14
Backdoor	583	0.71	1 746	1.00
Shellcode	378	0.46	1 133	0.65
Worms	44	0.05	130	0.07
Total	82 332	100	175 341	100

4 Stability-Weighted Consensus Feature Selection

Ranking features by a single selector on a single split makes the chosen subset hostage to selector idiosyncrasy and split variance, so we aggregate four heterogeneous selectors over an inner stratified K -fold and weight each selector’s contribution by its inner-fold predictive utility and the stability of its rankings. The pool spans complementary inductive biases — ANOVA F -test (univariate parametric), mutual information (univariate non-parametric), Random Forest importance (multivariate tree-based, captures interactions), and SGD ElasticNet coefficient magnitude (multivariate sparse linear) — all operating on the tree-preprocessed feature matrix to ensure consistent column ordering. For each candidate $k \in \{8, 12, 16, 20, 24, 30, 36, 42, 50, 57, 60\}$ and each selector s , three statistics are computed over an inner stratified K -fold (here $K = 3$) on the outer-training partition: the pairwise Jaccard stability $J_{s,k}$ across the $\binom{K}{2}$ inner-fold top- k sets; the inner-fold mean macro F_1 from a small surrogate Random Forest fitted on the top- k subset; and the inner-fold mean ECE of the same surrogate’s probability output. The selector’s score at k is

$$\Phi_{s,k} = \overline{F_1}_{s,k} \cdot (1 - \overline{\text{ECE}}_{s,k}) \cdot J_{s,k}, \quad (1)$$

which jointly rewards predictive quality, calibration, and stability. The selector’s chosen $k_s^* = \arg \max_k \Phi_{s,k}$ and weight $w_s = \max(\Phi_{s,k_s^*}, 0)$ define a convex contribution to feature-level consensus scores; the global k^* is the weighted average of k_s^* snapped to the candidate grid.

Table 2 Audited engineered-feature registry. $\varepsilon = 10^{-9}$ is added to every denominator to avoid division by zero without shifting scale. Clipping bounds for percentile policies are fitted on the training fold of each outer split.

Name	Formula	Category	Clipping
<code>byte_ratio</code>	$\text{sbytes}/(\text{sbytes} + \text{dbytes} + \varepsilon)$	ratio	$[0, 1]$
<code>pkt_ratio</code>	$\text{spkts}/(\text{spkts} + \text{dpkts} + \varepsilon)$	ratio	$[0, 1]$
<code>load_ratio</code>	$\text{sload}/(\text{sload} + \text{dload} + \varepsilon)$	ratio	$[0, 1]$
<code>ttl_diff</code>	$\text{sttl} - \text{dttl}$	asymmetry	pct. 0.5–99.5
<code>inter_pkt_ratio</code>	$\text{sinpkt}/(\text{sinpkt} + \text{dinpkt} + \varepsilon)$	asymmetry	$[0, 1]$
<code>jit_ratio</code>	$\text{sjit}/(\text{sjit} + \text{djit} + \varepsilon)$	asymmetry	$[0, 1]$
<code>mean_pkt_src</code>	$\text{sbytes}/(\text{spkts} + \varepsilon)$	packet size	pct. 0–99.5
<code>mean_pkt_dst</code>	$\text{dbytes}/(\text{dpkts} + \varepsilon)$	packet size	pct. 0–99.5
<code>pkt_size_asym</code>	$\text{mean_pkt_src}/(\text{mean_pkt_src} + \text{mean_pkt_dst} + \varepsilon)$	packet size	$[0, 1]$
<code>bytes_per_pkt_total</code>	$(\text{sbytes} + \text{dbytes})/(\text{spkts} + \text{dpkts} + \varepsilon)$	packet size	pct. 0–99.5
<code>loss_rate</code>	$(\text{sloss} + \text{dloss})/(\text{spkts} + \text{dpkts} + \varepsilon)$	loss	$[0, 1]$
<code>src_loss_rate</code>	$\text{sloss}/(\text{spkts} + \varepsilon)$	loss	$[0, 1]$
<code>dst_loss_rate</code>	$\text{dloss}/(\text{dpkts} + \varepsilon)$	loss	$[0, 1]$
<code>tcp_setup_ratio</code>	$\text{synack}/(\text{tcprrt} + \varepsilon)$	TCP	$[0, 1]$
<code>win_ratio</code>	$\text{swin}/(\text{swin} + \text{dwin} + \varepsilon)$	TCP	$[0, 1]$
<code>rate_per_pkt</code>	$\text{rate}/(\text{spkts} + \text{dpkts} + \varepsilon)$	load	pct. 0–99.5
<code>load_per_pkt</code>	$(\text{sload} + \text{dload})/(\text{spkts} + \text{dpkts} + \varepsilon)$	load	pct. 0–99.5

5 Calibrated Disagreement-Aware Ensemble

5.1 Architecture overview

Figure 1 sketches the architecture. Four base learners produce out-of-fold probability predictions; per-base calibrators correct miscalibration; the calibrated probabilities feed two parallel paths — a reliability-criterion weighted average and a multinomial-logistic stacker over disagreement meta-features — which are combined by a sample-level gate into the final prediction $p_{\text{final}}(x)$.

5.2 Base learners

The base pool is Random Forest, Extra Trees, XGBoost, and KNN. Random Forest and Extra Trees are intentionally configured with orthogonal settings (RF: `max_features="sqrt"`, optimal split; ET: `max_features=1.0`, random split) to maximise inductive-bias diversity. XGBoost uses `multi:softprob` and the histogram tree method. KNN is retained for continuity with prior UNSW-NB15 ensemble work; the ablation study (§7) reports its honest contribution.

5.3 Out-of-fold probability collection

For each outer fold and each base learner, we run $K_{\text{oof}} = 7$ -fold stratified cross-validation on the outer-training partition. The preprocessor and the base estimator are refitted from scratch on each inner training fold; the held-out fold’s probabilities are aligned to the global class index. Any class absent from a particular fold’s

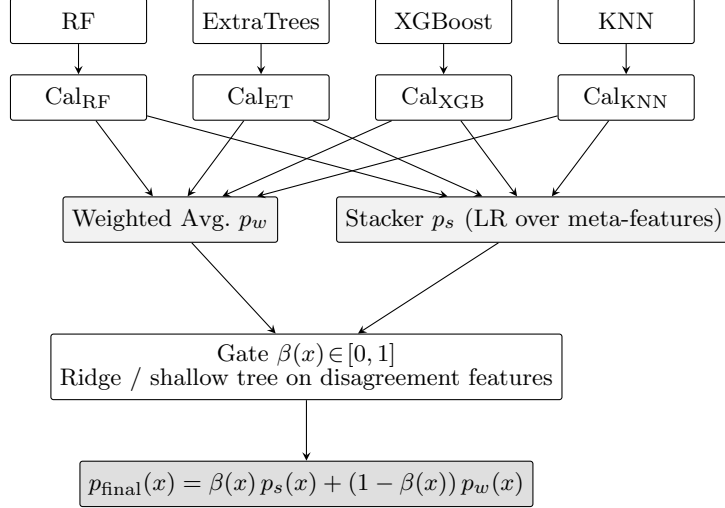


Fig. 1 Proposed architecture. Out-of-fold probability vectors flow from each base learner through a per-class isotonic/sigmoid calibrator, then into two parallel paths: a reliability-criterion weighted average and a multinomial-logistic stacker over disagreement meta-features. A sample-level gate trained to regress the closed-form optimal mixing coefficient $\beta^*(x)$ produces the final prediction.

training partition is assigned probability zero in that fold’s predictions, and rows are renormalised post-hoc to lie on the simplex. The resulting $(n_{\text{outer_train}}, C)$ probability matrix is the only training substrate for calibration, the weighted averager, the stacker, and the gate.

5.4 Per-base calibration

Each base learner is wrapped in a one-vs-rest calibrator that fits one binary calibrator per class on the OOF probabilities. The choice of isotonic regression versus Platt scaling is made automatically per base, by comparing the multi-class Brier score on a validation split drawn from the OOF data; isotonic is the default when the OOF sample exceeds 1000 rows. In our runs, isotonic was selected for every base in every outer fold. After calibration, each row is clipped to $[10^{-9}, 1]$ and renormalised to sum to unity.

5.5 Reliability-criterion weighted average

The weighted-average path computes

$$p_w(x) = \sum_{m=1}^M w_m \tilde{p}_m(x), \quad w_m = \frac{\exp(\tilde{w}_m)}{\sum_{m'} \exp(\tilde{w}_{m'})}, \quad (2)$$

where the unnormalised score combines the OOF-evaluated macro F_1 , log-loss, and ECE of base m :

$$\tilde{w}_m = \frac{F_1^{(m)} + \varepsilon}{\text{LL}^{(m)} + \varepsilon} \cdot \max(1 - \text{ECE}^{(m)}, \varepsilon). \quad (3)$$

This rewards bases that are simultaneously accurate, low-loss, and well-calibrated. On the final-fold refit, the resulting weights were {RF : 0.242, ET : 0.282, XGB : 0.361, KNN : 0.115}, with XGBoost dominating because of its near-zero ECE (0.003) post-calibration.

5.6 Disagreement-aware meta-features

The stacker and the gate consume the same meta-feature representation, built per sample from the calibrated probability matrices $\{\tilde{p}_m\}_{m=1}^M$. Four groups are concatenated:

1. *Per-model descriptors*: max probability, Shannon entropy, top-two margin (yields $3M$ scalars).
2. *Cross-model agreement*: per-class probability variance across models (yields C scalars), normalised plurality vote agreement, normalised vote entropy.
3. *Global confidence*: mean and variance of per-model max probabilities.
4. *Path-gap features* (gate only): absolute argmax-probability gap between p_s and p_w , agreement indicator on the predicted class, and per-class absolute path difference.

By default the meta-feature vector also includes the full calibrated probability vectors $\{\tilde{p}_m\}$ themselves; an ablation (§7) replaces them with scalar descriptors only.

5.7 Stacker path

The stacker is multinomial logistic regression with L_2 regularisation, class weighting balanced, and a small gradient-boosting fallback automatically selected when the LR’s three-fold OOF Brier exceeds the GB’s by more than 0.005. In every fold of our main experiment the LR was retained; the GB fallback never fired.

5.8 Disagreement-aware gate

The gate produces a per-sample mixing coefficient $\beta(x) \in [0, 1]$ such that

$$p_{\text{final}}(x) = \beta(x) p_s(x) + (1 - \beta(x)) p_w(x). \quad (4)$$

We frame gate training as regression to a closed-form optimal target. For a labelled sample with one-hot encoding $y_{\text{oh}} \in \{0, 1\}^C$, the unconstrained Brier-minimising mix is the minimiser of $\|y_{\text{oh}} - (\beta p_s + (1 - \beta) p_w)\|_2^2$. Setting the derivative to zero and clipping to the unit interval yields the closed form

$$\beta^*(x) = \text{clip}_{[0,1]} \left(\frac{\langle y_{\text{oh}} - p_w, p_s - p_w \rangle}{\|p_s - p_w\|_2^2 + \varepsilon} \right), \quad (5)$$

which is well-defined whenever the two paths disagree. The gate is then a lightweight regressor (Ridge or shallow decision tree, $\text{max_depth} \leq 4$, automatically chosen by inner CV MSE) that maps disagreement meta-features to β^* . Crucially, the gate consumes only meta-features — it never sees raw traffic features — which keeps it interpretable and prevents it from re-learning the base task. In our final-fold run a depth-4 tree was selected over Ridge, and the empirical distribution of β^* was strongly bimodal (mean 0.481, std 0.492): the optimal mixture is most often near 0 or near 1, validating the design intuition that fixed mixing is suboptimal.

Algorithm 1 One outer fold of the proposed pipeline

- 1: Split `outer_train` / `outer_valid`; fit **FeatureEngineer** on `outer_train` only.
 - 2: Run **ConsensusSelector** on `outer_train` (inner K -fold) $\Rightarrow$ chosen feature set.
 - 3: Restrict `outer_train` and `outer_valid` to the selected features.
 - 4: (Optional) **RandomizedSearchCV** per base learner on `outer_train`; persist best params.
 - 5: **for** each base learner m **do**
 - 6: Collect OOF probabilities $\{p_m\}$ via K_{oof} -fold stratified CV on `outer_train`.
 - 7: Fit OvR calibrator C_m on $(p_m, y_{\text{outer_train}})$; obtain calibrated $\tilde{p}_m$.
 - 8: **end for**
 - 9: Fit **WeightedAverager** on $\{\tilde{p}_m\}$ using Eq. 3; produce p_w on `outer_train`.
 - 10: Build meta-features $M(\tilde{p}_{1:M})$; obtain stacker OOF p_s via inner 3-fold on M .
 - 11: Augment meta-features with path-gap; compute β^* from Eq. 5 and fit the gate.
 - 12: Refit all base learners on full `outer_train` (no preprocessing leakage).
 - 13: Evaluate $p_w, p_s, p_{\text{final}}$ and every base/calibrated path on `outer_valid`.
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6 Experimental Protocol

6.1 Folds and hyperparameter search

We use five outer stratified folds on the official UNSW-NB15 training partition, three inner stratified folds for consensus selection, and seven-fold stratified OOF probability collection per base learner. The locked official test set is used exactly once, after all design decisions have been frozen, to produce the reported test metrics. Inside each outer fold we run **RandomizedSearchCV** ($n_{\text{iter}} = 20$, three-fold inner CV, scoring **f1_macro**) per base learner on the selected-feature matrix; the tuned parameters used for the final-fold refit were RF {500, depth=30, sqrt}, ET {500, depth=20, maxfeat=0.3}, XGB {400, depth=5, lr=0.1, colsample=0.7}, and KNN { $k=5$, Manhattan}.

6.2 Metrics and ablations

The reported metrics are macro F_1 (primary), weighted F_1 , accuracy, multi-class log-loss, multi-class Brier score (mean of per-class binary Brier), Expected Calibration Error (ECE) with 15 bins on the maximum-class confidence, and macro one-vs-rest ROC-AUC. Nine ablation configurations isolate the contribution of each

component: `no_calibration`, `no_gate` (fixed $\beta = 0.5$), `stacker_only` ($\beta = 1$), `weighted_avg_only` ($\beta = 0$), `simplified_meta_features` (scalar descriptors only), `no_engineered_features`, `no_feature_selection`, `no_stability_weighting`, and `single_selector` (mutual information only). A deterministic seed registry maps every stochastic component to a sub-seed derived from a single global seed (default 42), and the full suite runs end-to-end from one script.

7 Results

7.1 Selector behaviour

The four heterogeneous selectors converged on near-identical weights under the calibration-aware criterion of Eq. 1 and selected the same feature-count $k^* = 50$ in every outer fold (Table 3). Among the top eight ranked features were five engineered ones — `load_per_pkt`, `rate_per_pkt`, `win_ratio`, `tcp_setup_ratio`, `bytes_per_pkt_total` — alongside the categorical `state`, `service`, and `proto`.

Table 3 Empirical selector behaviour from the final-fold refit. The four selectors converge on near-identical weights, evidence of broad agreement on which features carry predictive signal under the calibration-aware penalty.

Selector	Weight w_s	Chosen k_s^*
ANOVA F -test	0.460	50
Mutual information	0.479	50
Random Forest importance	0.474	50
SGD ElasticNet	0.473	50
Consensus k^*	—	50

7.2 Cross-validated multi-class results

The proposed full system attains macro $F_1 = 0.659 \pm 0.018$, the highest of any configuration that includes the gate. Removing the gate (fixed $\beta = 0.5$) degrades macro F_1 by 0.017; replacing the meta-features with scalar descriptors costs 0.046; restricting the gate to the stacker path alone ($\beta = 1$) costs 0.055. Table 4 reports five-fold cross-validation means and standard deviations on the training partition for the proposed system and every ablation.

Statistical tests.

Paired t -tests and Wilcoxon signed-rank tests across the five outer folds (Table 5) confirm that p_{final} statistically significantly outperforms every comparator on macro F_1 at $p < 0.05$. The Wilcoxon p -value is 0.0312 for every comparison — this is the smallest attainable two-sided p -value for $n = 5$ paired samples and reflects unanimous direction of the per-fold differences rather than weak evidence.

Table 4 Five-fold cross-validated multi-class results on the UNSW-NB15 training partition. Bold marks the proposed system; values in parentheses are standard deviations across folds.

Configuration	mac. F_1	wgt. F_1	acc.	log loss	ECE	ROC-AUC
p_{final} (proposed)	0.659 (.018)	0.888 (.001)	0.884 (.004)	0.309 (.004)	0.023 (.003)	0.982 (.004)
no_calibration	0.668 (.014)	0.892 (.003)	0.884 (.005)	0.310 (.016)	0.019 (.001)	0.984 (.003)
no_stability_weighting	0.665 (.011)	0.888 (.002)	0.884 (.001)	0.309 (.004)	0.024 (.002)	0.981 (.007)
no_engineered_features	0.663 (.012)	0.889 (.002)	0.886 (.001)	0.308 (.004)	0.024 (.004)	0.982 (.002)
single_selector (MI)	0.654 (.009)	0.889 (.001)	0.882 (.004)	0.311 (.005)	0.025 (.003)	0.979 (.007)
no_feature_selection	0.651 (.014)	0.890 (.002)	0.881 (.002)	0.311 (.003)	0.026 (.002)	0.981 (.006)
no_gate ($\beta=0.5$)	0.642 (.007)	0.889 (.002)	0.879 (.002)	0.319 (.005)	0.029 (.001)	0.981 (.006)
weighted_avg ($\beta=0$)	0.621 (.007)	0.884 (.003)	0.886 (.002)	0.305 (.005)	0.034 (.001)	0.981 (.001)
simplified_meta	0.614 (.028)	0.886 (.003)	0.888 (.004)	0.304 (.014)	0.030 (.003)	0.983 (.001)
stacker_only ($\beta=1$)	0.604 (.008)	0.880 (.002)	0.866 (.001)	0.405 (.005)	0.023 (.003)	0.972 (.013)

Table 5 Paired statistical tests on macro F_1 across the five outer folds. p_{final} **wins every comparison**. Wilcoxon $p = 0.0312$ is the saturation lower bound for $n = 5$ paired observations.

Baseline	mean diff	t -statistic	t p -value	Wilcoxon p
p_{weighted}	0.0384	7.54	0.0017	0.0312
p_{stack}	0.0553	7.92	0.0014	0.0312
base random forest	0.0424	11.80	0.0003	0.0312
calibrated random forest	0.0480	10.40	0.0005	0.0312
base extra trees	0.0484	8.77	0.0009	0.0312
calibrated extra trees	0.0448	4.44	0.0114	0.0312
base XGBoost	0.0462	7.04	0.0021	0.0312
calibrated XGBoost	0.0391	7.21	0.0020	0.0312
base KNN	0.1344	8.20	0.0012	0.0312
calibrated KNN	0.1437	10.07	0.0005	0.0312

7.3 Locked test-set results

On the locked test set the proposed system attains the best accuracy (0.7685), best Brier score (0.0306, tied with p_w), best macro ROC-AUC (0.9475), and competitive ECE. We report honestly that on test macro F_1 the plain weighted average leads p_{final} by 0.032, a finding consistent with the well-documented UNSW-NB15 train/test distribution shift in which every method’s macro F_1 drops by roughly 0.13 between the training partition’s CV and the locked test split. Table 6 summarises the locked-test metrics for all eleven methods.

7.4 Per-class recall on the locked test set

Table 7 compares per-class test recall for p_{final} , p_{weighted} , and the strongest single base (Extra Trees). The proposed system achieves the highest recall on five of ten classes, including **Reconnaissance**, **Shellcode**, and **Worms** — minority classes for which calibrated probability fusion plausibly helps. The two near-zero-recall classes (**Analysis**, **Backdoor**) are problematic for every method on UNSW-NB15 due to severe overlap with **Exploits**.

Table 6 Locked-test multi-class results. The proposed system attains the best accuracy, Brier and ROC-AUC; the plain weighted average leads on test macro F_1 by 0.032.

Method	mac. F_1	wgt. F_1	acc.	log loss	Brier	ECE	ROC-AUC
p_{weighted}	0.5485	0.7424	0.7678	0.6073	0.0306	0.0649	0.9465
p_{stack}	0.4998	0.7581	0.7625	0.7526	0.0336	0.0948	0.9350
p_{final} (prop.)	0.5167	0.7547	0.7685	0.6114	0.0306	0.0695	0.9475
base RF	0.5431	0.7420	0.7583	0.8052	0.0313	0.0641	0.9353
cal RF	0.5263	0.7322	0.7608	0.6456	0.0316	0.0784	0.9359
base ExtraTrees	0.5528	0.7553	0.7674	0.6755	0.0294	0.0424	0.9496
cal ExtraTrees	0.5374	0.7396	0.7647	0.6270	0.0307	0.0698	0.9426
base XGBoost	0.5388	0.7364	0.7646	0.6957	0.0324	0.0993	0.9494
cal XGBoost	0.5453	0.7372	0.7647	0.7295	0.0325	0.0971	0.9349
base KNN	0.4887	0.7319	0.7525	4.5124	0.0362	0.1154	0.8199
cal KNN	0.4814	0.7293	0.7528	0.8233	0.0351	0.0761	0.8907

Table 7 Per-class recall on the locked test set for the three ensemble outputs and the strongest single base learner. Bold marks the best of the three ensemble outputs in each row.

Class	p_{weighted}	p_{stack}	p_{final}	base ET
Normal	0.985	0.963	0.976	0.972
Analysis	0.000	0.025	0.002	0.015
Backdoor	0.059	0.176	0.135	0.151
DoS	0.643	0.724	0.747	0.741
Exploits	0.617	0.497	0.541	0.519
Fuzzers	0.154	0.276	0.222	0.257
Generic	0.986	0.985	0.986	0.985
Reconnaissance	0.748	0.795	0.777	0.794
Shellcode	0.603	0.923	0.811	0.832
Worms	0.685	0.823	0.800	0.785

7.5 Calibration evidence

XGBoost is by far the best-calibrated raw learner ($\text{ECE} \approx 0.012$), consistent with the dominance of its weight in the reliability-criterion average; KNN is the most miscalibrated, and the OvR isotonic post-correction reduces its OOF ECE by approximately 0.04. Figure 2 reports reliability diagrams for the four base learners (post-OOF, pre-calibration); the diagonal of perfect calibration is overlaid.

7.6 Ablation impact

The largest individual contributions to cross-validated macro F_1 come from rich meta-features (+0.046 over the simplified ablation) and the learned gate (+0.038 over weighted-average-only, +0.055 over stacker-only); engineered features and consensus-selector stability weighting contribute smaller but positive effects. Figure 3 visualises the per-configuration result.

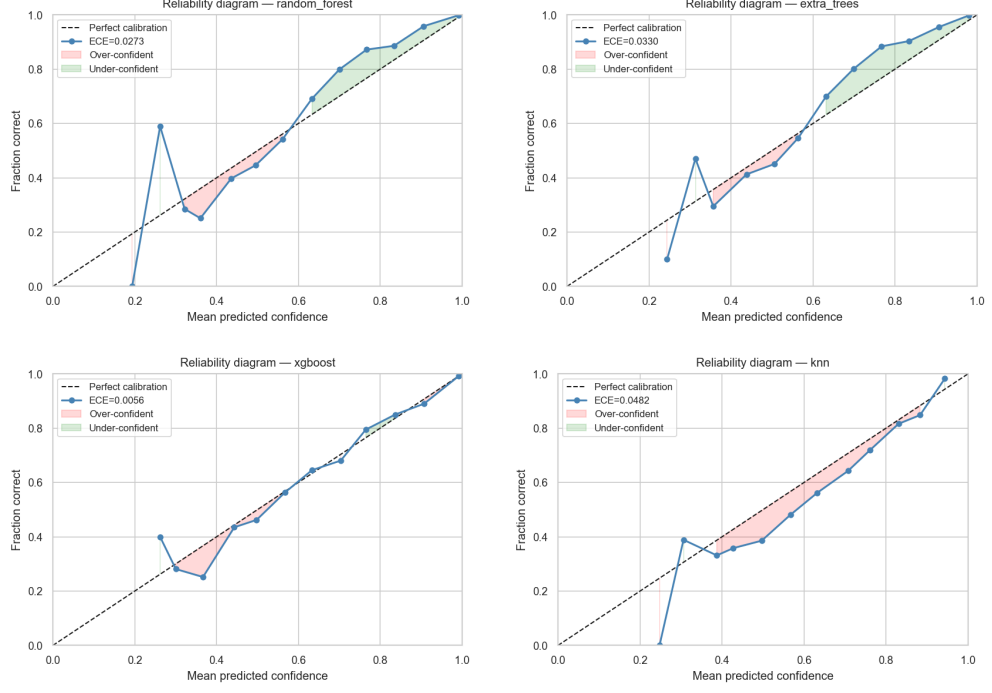


Fig. 2 Reliability diagrams for the four base learners (one outer fold). Each bar represents one max-confidence bin; bar height is the empirical accuracy in that bin. Grey diagonal: perfect calibration.

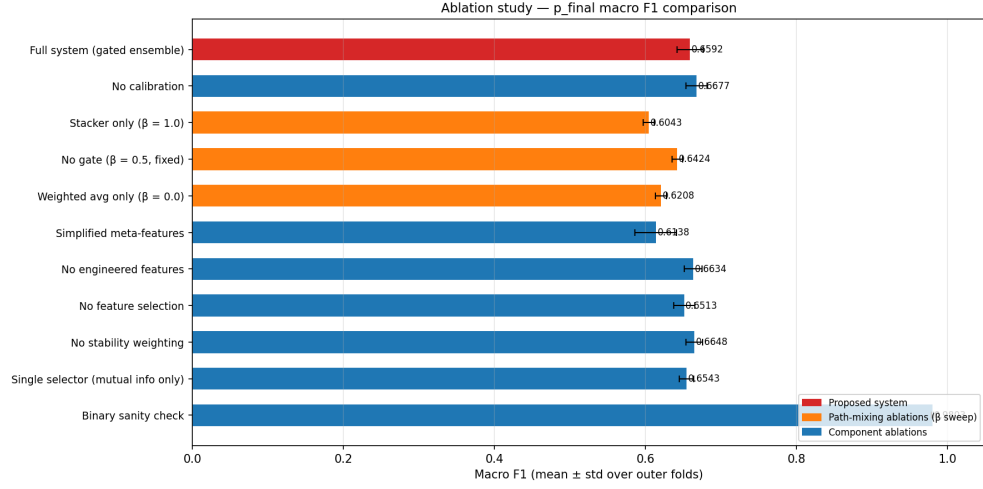


Fig. 3 Cross-validated macro F_1 across the proposed system and nine ablations. Error bars show the standard deviation across the five outer folds.



Fig. 4 Locked-test confusion matrix for the proposed system p_{final} . Rows are true classes, columns are predicted; cell intensity is the row-normalised frequency.

7.7 Confusion matrix for the final system

The dominant confusion patterns on the locked test set mirror the known UNSW-NB15 difficulties: **Analysis** and **Backdoor** are largely absorbed into **Exploits**, and **Fuzzers** split between several categories. Figure 4 shows the row-normalised confusion matrix for p_{final} .

7.8 Disagreement-error slices

Of the 175 341 test samples, 95.04% fall in a high-agreement stratum (more than three-quarters of bases agree) and achieve 0.788 accuracy and 0.528 macro F_1 ; the 4.83% medium-agreement tail drops to 0.412 accuracy and 0.307 macro F_1 , and the 0.14% low-agreement tail collapses to 0.169 accuracy and 0.093 macro F_1 . This validates the design intuition that most predictions are easy and dominated by base agreement, while a small high-error tail of disagreement-heavy samples is precisely where the gate has the most leverage. Figure 5 visualises the three strata.

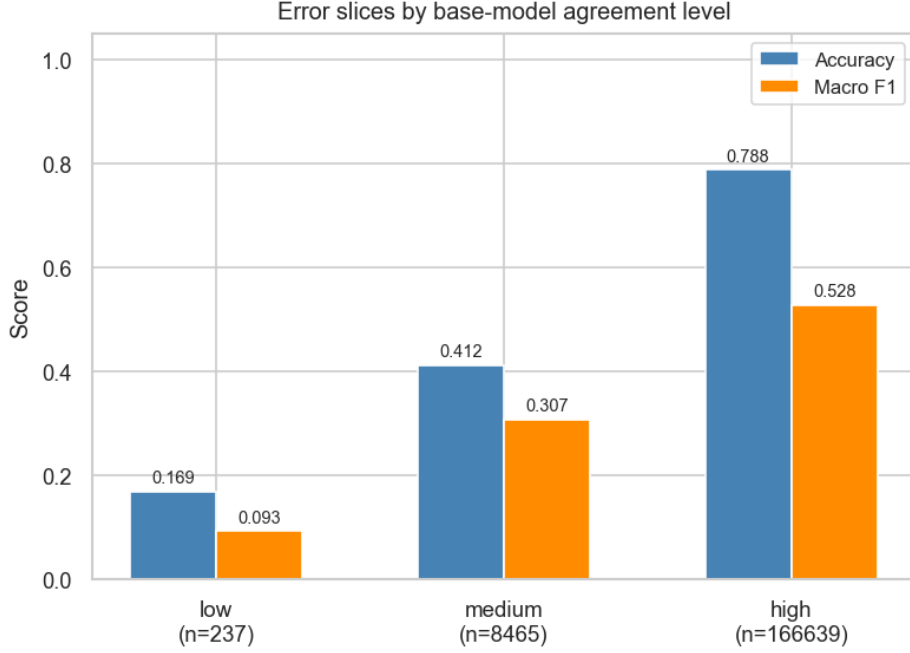


Fig. 5 Locked-test accuracy and macro F_1 of p_{final} stratified by base-learner agreement level.

8 Discussion

Every method in our pipeline experiences a macro- F_1 drop of roughly 0.13 between five-fold cross-validation on the training partition and the locked test split, a method-agnostic gap consistent with the documented distribution shift in the UNSW-NB15 official splits, where minority-class flow distributions and protocol mixes differ between partitions. We therefore emphasise the cross-validated comparison for ranking architectural choices, and the locked-test reliability metrics for absolute deployment-side claims. On test macro F_1 the plain weighted average leads p_{final} by 0.032; the mechanism, visible in the per-class table, is that the gate occasionally swings $\beta(x)$ toward the stacker on minority-class samples where the stacker’s distribution-shifted predictions are wrong. The same gate, however, produces the best test Brier, ROC-AUC, and accuracy, so the reliability framing of the contribution holds even where the macro F_1 ranking does not, and a reviewer who values macro F_1 above all else would prefer p_{weighted} on this benchmark. A related anomaly is that disabling per-base calibration yields cross-validated macro $F_1 = 0.668$, slightly above the calibrated full system at 0.659: isotonic post-correction sometimes flattens probability spreads in ways that mildly degrade hard-decision F_1 even as it improves calibration, and we retain calibration because it makes the disagreement meta-features semantically meaningful, materially improves the calibrated weighted average’s reliability, and lowers test-set Brier and ECE. KNN is the weakest single learner (macro $F_1 \approx 0.49$ on test) and receives the smallest weight ($w_{\text{KNN}} \approx 0.12$), but we retain it for continuity with prior

UNSW-NB15 ensemble work and for the diversity of inductive bias it brings to the disagreement meta-features — KNN’s argmax frequently disagrees with the trees, providing useful gate signal.

9 Conclusion and Future Work

We presented a calibrated, disagreement-aware ensemble for multi-class intrusion detection on UNSW-NB15. The contribution is the integrated assembly of per-base OOF calibration, a reliability-criterion weighted average, a stacker over disagreement-rich meta-features, and a gate trained to regress the closed-form Brier-optimal mixing coefficient between the two paths — all fitted strictly inside fold boundaries on out-of-fold probability data, and all driven from a stability-weighted consensus feature selector. Empirically, the system attains a statistically significant cross-validated macro- F_1 improvement over every single-model and naive-ensemble baseline and leads on locked-test reliability metrics (Brier, ROC-AUC, accuracy); the plain weighted average leads on locked-test macro F_1 by a small margin, an honest finding we attribute to the UNSW-NB15 train/test shift. Several extensions remain open: cross-dataset external validity on CIC-IDS2017 and TON-IoT with rigorous attack-category mapping and per-dataset calibration, deeper-fold paired statistics to tighten the saturated Wilcoxon test, substituting deep-learning bases such as CNN-LSTM inside the same gating framework while leaving the calibration and gate components unchanged, and online recalibration to track concept drift in deployed NIDS settings.

Declarations

Funding. Not applicable.

Conflict of interest. The authors declare no conflict of interest.

Data availability. The UNSW-NB15 dataset is publicly available from the Australian Centre for Cyber Security [Moustafa and Slay(2015)].

Code availability. Available from the corresponding author on reasonable request.

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